---

📘 **Assignment: Microservices with API Gateway**

✈️ **Domain: Flight Management System**

---

🎯 **Objective**

Design and implement a **basic microservices architecture** with:

- Multiple **dummy microservices**
- An **API Gateway** (using Ocelot or YARP)
- Proper **routing and communication**

---

🧩 **Problem Statement**

You are building a **Flight Management System** for an airline company.

Instead of a monolithic system, you must:

- Split functionality into **independent microservices**
- Expose them through a **single API Gateway**

---

📐 **Architecture Overview**

```
Client (Postman / Angular App)
        ↓
    API Gateway
        ↓
┌──────────┼──────────┐
↓        ↓        ↓
Flight   Booking   Passenger
Service  Service   Service
```

---

## 📦 Microservices to Create (Dummy APIs)

### 1️⃣ Flight Service

**Responsibilities:**

- Manage flight details

**Endpoints:**

- GET /api/flights
- GET /api/flights/{id}
- POST /api/flights

**Sample Response:**

```
{
  "id": 1,
  "flightNumber": "AI101",
  "source": "Hyderabad",
  "destination": "Delhi"
}
```

---

### 2️⃣ Booking Service

**Responsibilities:**

- Manage bookings

**Endpoints:**

- GET /api/bookings
- POST /api/bookings

**Sample Response:**

```
{
  "bookingId": 101,
```

"flightId": 1,

"passengerId": 201

}

---

### 3️⃣ Passenger Service

**Responsibilities:**

- Manage passenger details

**Endpoints:**

- GET /api/passengers
- POST /api/passengers

**Sample Response:**

{

"passengerId": 201,

"name": "John Doe",

"age": 30

}

---

### 🌐 API Gateway Implementation

Use :

- Ocelot

---

### 🔧 Gateway Responsibilities

- Route requests to respective microservices
- Act as a **single entry point**
- Handle URL mapping

---

## ⚙️ Implementation Steps

**Step 1: Create Solutions**

- Create **separate ASP.NET Core Web APIs**:
  - FlightService
  - BookingService
  - PassengerService
  - API Gateway

---

**Step 5: Test via Gateway**

Use Postman:

| Service | Direct URL | Gateway URL |
|---|---|---|
| Flight | /api/flights | /gateway/flights |
| Booking | /api/bookings | /gateway/bookings |

---

## 🧪 Expected Output

When calling:

GET http://localhost:5000/gateway/flights

➡️ Response should come from **Flight Service via Gateway**